Reading 1.1 Ship Classification

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The problem: one ship, many labels

There is no single, universal definition of ship types; IMO itself says so. Each community that deals with ships uses its own vocabulary, so the same ship, with the same IMO number, hull and cargo, can carry four different labels:

Source	Label for the same ferry
A commercial ship database	Passenger vessel
EU MRV reporting	Ro-pax ship
MARPOL Annex VI, Chapter 4	Ro-ro passenger ship
A broker's fleet report	Passenger

None of these labels is wrong: each is correct within its own system. The difficulty is that the systems agree on the main groups, such as tankers or container ships, and split the sub-groups differently. The EU separates oil tankers from chemical tankers where other sources speak only of tankers; commercial databases put dredgers, tugs and port craft into miscellaneous groups. Anyone who compares, regulates or plans for ships from more than one source has to translate between them first.

IMO ship types: who defines what

IMO's definitions are spread across its instruments, and each instrument defines only the types it regulates, so the scope of the same name can differ from one to the next.

- **SOLAS** defines passenger ships and bulk carriers, for safety.
- **MARPOL Annex I** defines the oil tanker, for oil pollution; **Annex II** covers chemical (noxious liquid substance) tankers.
- **The MODU Code** defines mobile offshore drilling units; **the HSC Code** high-speed craft.
- **MARPOL Annex VI, Chapter 4** is the reference for decarbonisation rules. Its revised text, adopted by Resolution MEPC.328(76) in 2021, defines in Regulation 2.2 the ship types that must meet the EEDI, EEXI and CII requirements.

The main definitions, in summary:

Ship type	What defines it
Passenger ship	Carries more than 12 passengers
Cruise passenger ship	Passenger ship with no cargo deck, for passengers in overnight accommodation on a sea voyage
Ro-ro passenger ship	Passenger ship with ro-ro cargo spaces
Bulk carrier	Mainly for dry bulk; single deck, top-side and hopper side tanks; includes ore and combination carriers
Combination carrier	Can load full deadweight of either liquid or dry bulk cargo
Oil tanker	Mainly for oil in bulk; includes combination carriers, NLS tankers and gas carriers carrying oil
Tanker	An oil tanker (MARPOL Annex I) or a chemical/NLS tanker (Annex II)
Gas carrier	Carries liquefied gas in bulk, other than an LNG carrier
LNG carrier	Carries liquefied natural gas in bulk
Containership	Designed exclusively for containers, in holds and on deck
General cargo ship	Mainly for general cargo; for the EEDI reference line, excludes livestock, barge, heavy-load, yacht and nuclear fuel carriers
Refrigerated cargo carrier	Designed exclusively for refrigerated cargo in holds
Ro-ro cargo ship	Designed for roll-on/roll-off cargo units
Ro-ro cargo ship (vehicle carrier)	Multi-deck ro-ro ship for empty cars and trucks
Special purpose ship	Carries more than 12 special personnel by reason of its function (research, training, offshore wind maintenance)
Fishing vessel	Used to catch fish or other living resources of the sea
High-speed craft	Capable of a speed above a limit set by its displaced volume

Ship type	What defines it
Mobile offshore drilling unit	Able to drill for seabed resources such as oil, gas, sulphur or salt
Nuclear ship	Has a nuclear power plant

The list is not exhaustive. Passenger ships in general, fishing vessels, drilling units, special purpose ships and high-speed craft are defined outside Annex VI, in SOLAS, MARPOL Annex I or their own codes.

EU ship types: MRV, ETS and FuelEU

The EU classifies ships through one regulation and reuses that classification everywhere. The MRV Regulation, (EU) 2015/757, sets the ship types for monitoring and reporting emissions; EU ETS Maritime (Directive (EU) 2023/959) and FuelEU Maritime (Regulation (EU) 2023/1805) take over its scope instead of defining their own. A ship reported as a bulk carrier under MRV is a bulk carrier under ETS and FuelEU as well.

MRV names fourteen ship types, each with its own cargo measure (Implementing Regulation (EU) 2016/1928):

EU ship type	Cargo measure
Passenger ship	Number of passengers
Ro-pax ship	Passengers and mass of cargo
Ro-ro ship	Mass of cargo
Vehicle carrier	Mass of cargo
Container ship	Cargo mass, or TEU times a default mass
Container/ro-ro cargo ship	Cargo volume, from units, lane metres and TEU
General cargo ship	Deadweight carried on laden voyages
Refrigerated cargo carrier	Mass of cargo
Bulk carrier	Mass of cargo
Combination carrier	Mass of cargo
Oil tanker	Mass of cargo
Chemical tanker	Mass of cargo

EU ship type	Cargo measure
Gas carrier	Mass of cargo
LNG carrier	Volume of cargo discharged

Ships of 5,000 GT and above in commercial transport that fit none of these are reported as **other ship types**. FuelEU adds its own definitions of passenger ship, cruise passenger ship and containership, because it treats them differently for onshore power supply; the EU passenger ship is the one of Directive (EU) 2016/802, a ship carrying more than 12 passengers.

The EU list departs from IMO's in three places: it splits the tanker into oil tanker and chemical tanker, it keeps ro-pax ships apart from ro-ro ships, and it has separate types for container/ro-ro ships and vehicle carriers. It applies these choices consistently across MRV, ETS and FuelEU.

StatCode 5: the commercial and statistical standard

Outside the regulations, most ship data is coded with StatCode 5, the system of S&P Global (formerly IHS). It replaced the older International Classification of Ship Types (ICST) in 2009 and is now the working standard of commercial ship databases, shipbrokers, port systems and statistics offices, among them the UK Department for Transport.

StatCode 5 is descriptive, not legal: it says what a ship is, not which rules apply to it. Its strength is detail. Every ship is coded in five levels:

Level	Role	Example
1	Family	Cargo carrying
2	Group	Tankers
3	Sub-group	Liquefied gas
4	Ship type	LPG tanker
5	Detailed type	LPG/chemical tanker

Level 1 has six families: cargo carrying, work vessel, non merchant, non seagoing merchant, non propelled and non ship structure. The detail grows to about 300 types at Level 5. Levels 2 and 3 are where StatCode comes closest to the regulatory categories; below that, it distinguishes ships the rules treat as one type.

Industry size conventions by cargo

On top of the type, the market names ships by size. Each trade has its own convention and its own unit, chosen for what matters commercially, never gross tonnage:

Cargo	Convention	Classes, small to large	Unit
Oil and products	AFRA scale (London Tanker Brokers' Panel)	Handysize, MR, Panamax (LR1), Aframax, Suezmax, VLCC, ULCC	DWT
Dry bulk and combination carriers	Dry bulk market convention	Handysize, Handymax, Supramax, Ultramax, Panamax (Kamsarmax), Post-Panamax, Capesize, VLOC, Chinamax	DWT
Containers	Canal fit	Small feeder, feeder, feedermax, Panamax, Post-Panamax, Neo-Panamax, ULCV	TEU
Liquefied petroleum gas	LPG trade convention	MGC, LGC, VLGC, ULGC	m^3
Liquefied natural gas	Generations of the Qatar fleet	Small-scale, small conventional, conventional, large conventional, Q-Flex, Q-Max	m^3
Refrigerated cargo	Reefer trade convention	By hold capacity	cubic feet

The names are market labels: their limits move with the market and with the canals and ports that created them (Ship types 1.2).

Why the differences matter

As long as the systems stay apart, every use of ship data starts with a translation, and errors in that translation have real effects:

- **Compliance.** Under EU MRV, ETS and FuelEU, and IMO's CII, the ship type selects the reference line and the cargo measure. A ship placed in the wrong category is rated in the wrong band or follows the wrong compliance route; across a fleet and over a decade, the cost can run to millions.

- **Infrastructure planning.** A port planning shore power must know which ships it will serve. A list of "passenger ships" mixes cruise ships, which stay for days and need high power, with ferries, which turn round in hours and need fast, frequent connections.
- **Benchmarking.** A fleet's emissions intensity can only be compared with an industry benchmark if both use the same classification.
- **Comparing studies.** Studies of fleet power demand or emissions that start from different sources may not even be counting the same ships, so their results cannot be set side by side.
- **Hidden segments.** Where dredgers, tugs and port craft are pooled in miscellaneous groups, their very different operating profiles disappear from the data.

This is where the topic stands today: each system works within its own community, and the gap between them is managed case by case, by whoever needs to combine them.

Sources

- IMO: SOLAS; MARPOL Annexes I, II and VI (Annex VI as revised by Resolution MEPC.328(76), Regulation 2.2); MODU Code; HSC Code. Summary of the Annex VI ship types and thresholds: MGN 683, IMO carbon reduction measures.
- EU: Regulation (EU) 2015/757 (MRV); Implementing Regulation (EU) 2016/1928 (cargo carried); Directive (EU) 2023/959 (EU ETS, maritime); Regulation (EU) 2023/1805 (FuelEU Maritime); Directive (EU) 2016/802, Article 2(i) (passenger ship).
- S&P Global Market Intelligence (2023), *StatCode 5 Shiptype Coding System*.
- UK Department for Transport, Shipping fleet statistics: notes and definitions (StatCode 5 since 2009, ICST before).
- Papanikolaou, A. (2026), *Ship Design: Methodologies of Preliminary Design*, §1.3.6.2 (ship categories).